XRPL Validator — Daily Engineering Report · 2026-10-31

Period: 2026-10-31 00:00 to 24:00 US Eastern (2026-10-31 04:00 to 2026-11-01 04:00 UTC); ledgers #108,015,364 to #108,037,404. Every count from the ledger and the validator's logs stops at #108,037,404, closed 03:59:52 UTC. Generated 2026-11-01 00:20 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Status

Normal — coverage outside planned maintenance 100.000% (objective 99.9%), every validation agreed with the network, no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,041 of the 22,041 ledgers the network closed (100.000%); every validation matched the network's validated ledger.
- **Native engine:** 0 engine receipts.
- **Changes:** 0 deployments, 0 commits merged into main.
- **Network:** 1 slow-close period; the longest, 07:53 UTC–07:53 UTC, closed ledgers up to 19 s apart.
- **Independence program:** 3 of 30 qualifying days (snapshot at 04:20:44 UTC).
- **Amendment fixCleanup3_4_0** holds majority; activation due ~2026-11-04 14:46 UTC.

Service levels

Measure	Value
Ledgers closed by the network	22,041
Validations issued	22,041
Coverage (validations / ledgers closed)	100.000%
Validations matching the network's validated ledger hash	22,041 of 22,041
Unsigned during planned restarts	0
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%
Restarts	0
Basis: our own ledger hash	22,034
Basis: our own hash, witnessed by the reference node (quorum short)	7

Measure	Value
External agreement, data.xrpl.org (UTC day 2026-10-31)	1.00000 (0 missed of 21,126)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#108,015,364–#108,037,404), by step:

Step	Ledgers with a receipt
Step 1 · native engine	0
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	7
Step 5 · own hash	0
Writer	0

Live counters, snapshot at 04:20:44 UTC (since the last restart, 2026-10-28 22:52 UTC; not cut off at the period's last ledger):

Step	Checked	Result
1 · native engine vs libxrpl	69,560 ledgers, 13,276,246 transactions	69,559 fully matched; results 2, bytes 1, keys 0/0
2 · transaction records	70,011 ledgers, 13,509,783 records	13,509,780 exact; differ 3, tree 3
3 · transaction order	70,050 ledgers (mode: on)	70,048 exact; order 0, results 2, bytes 0, refused 0
4 · ledger data from peers	70,098 ledgers fetched	0 matched; differ 0, failed 0
4 · trusted validations	70,095 ledgers judged	70,088 agreed; differ 0, short of quorum 7
5 · our own ledger hash	70,098 ledgers	70,006 own; differ 3, partial 89
Writer	every ledger written	mismatch 1, distrusted 0

"Partial" ledgers are those the engine could not yet build alone while warming up after a restart.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	07:53 UTC	#108,018,954–#108,018,960 (7)	Step 4 · validations	16–27 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-10-31-01, INC-01 is INC-2026-10-31-01.

Incidents

No incidents.

Change log

Deployments

No deployments.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Independence program

Qualifying days: **3 of 30** (snapshot at 04:20:44 UTC). A UTC day qualifies when every validation is of a ledger our own engine built, with our own transaction order and ledger data taken from peers, and coverage is at least 98%.

UTC day 2026-10-31 — tracked #108,011,718–#108,033,757	Value
Validations	22,038 (22,038 own, 0 borrowed)
Coverage of the tracked ledgers	99.99%
Our own transaction order	on
Ledger data from peers	yes

Meets every criterion.

The validator saves this record every 64 validations, so a day in progress lags the signing log slightly; it is not cut off at the period's last ledger.

Network

Measure	Value
Ledgers closed	22,041 (#108,015,364–#108,037,404)
Mean close interval	3.92 s
Longest close interval	19 s (before #108,018,959)
Closes at least 15 s apart	1
Empty ledgers	0

- Slow closes 07:53:31 UTC–07:53:50 UTC: ledgers #108,018,959, 1 close at least 15 s apart, up to 19 s.

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-21 14:46 UTC	2026-11-04 14:46 UTC

Amendments enabled on mainnet: 97.

Open items

- Committed, awaiting merge and deployment: `c7d3f67` fix(the validator's ledger feed): the FFI leg skips pseudo-txs from peers too.
- Amendment fixCleanup3_4_0: activation due ~2026-11-04 14:46 UTC.

Corrections to earlier reports

None.